

Step	Reason
1) $p \wedge s$	Premise
2) p	Simplification on (1)
3) s	Simplification on (1)
4) $s \rightarrow \neg r$	Premise
5) $\neg r$	Modus ponens on (3), (4)
6) $p \rightarrow (q \vee r)$	Premise
7) $q \vee r$	Modus ponens on (2), (6)
8) $r \vee q$	Commutative law on (7)
9) q	Disjunctive syllogism on (8), (5)

Example 9: Show that the following argument is valid. If Mohan is a lawyer, then he is ambitious. If Mohan is an early riser, then he does not like idlies. If Mohan is ambitious, then he is an early riser. Then, if Mohan is a lawyer, he does not like idlies.

Solution: L: Mohan is a lawyer A: Mohan is ambitious
R: Mohan is an early riser I: Mohan likes idlies.

Premises: 1) $L \rightarrow A$ 2) $R \rightarrow \neg I$ 3) $A \rightarrow R$

Conclusion: $L \rightarrow \neg I$

Step	Reason
1) $A \rightarrow R$	Premise
2) $R \rightarrow \neg I$	Premise
3) $A \rightarrow \neg I$	Hypothetical syllogism on (1), (2)
4) $L \rightarrow A$	Premise
5) $L \rightarrow \neg I$	Hypothetical syllogism on (4), (3)